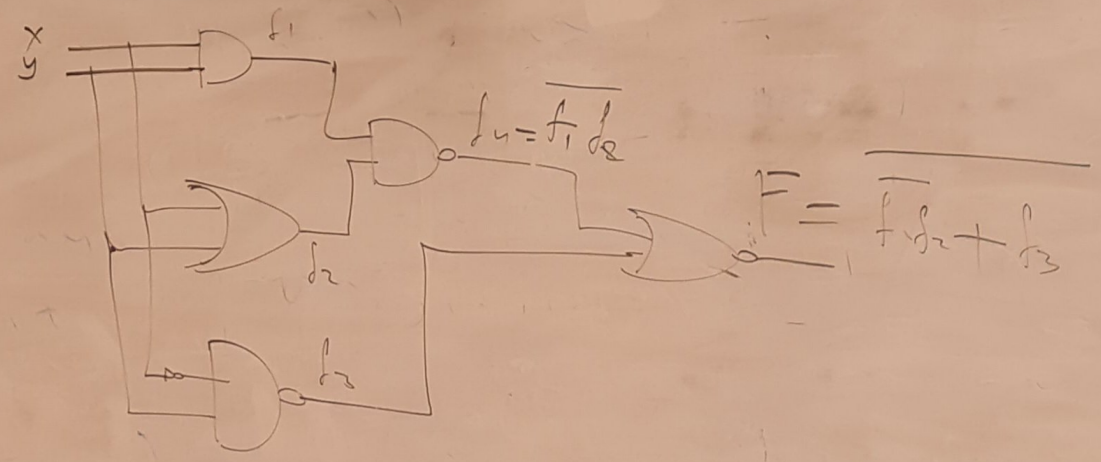


12

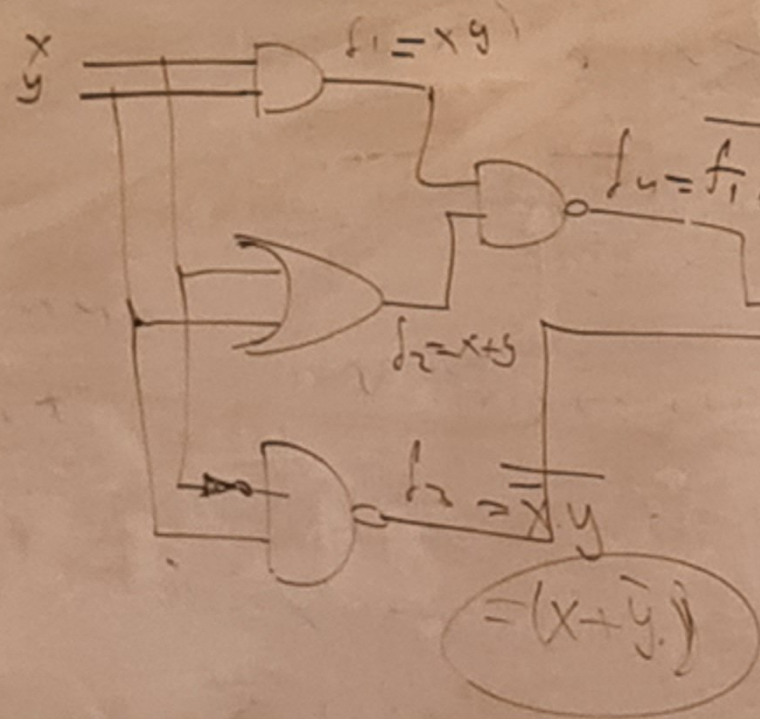
Active and Passive Logic

Analysis of comb. logic (F)



Active and Passive Wire

Logic (F)?!



$$\begin{aligned} &= xy(x+y) = \overline{xy} + \overline{(x+y)} \\ &= \overline{xy} + \overline{x+y} \end{aligned}$$

$$F = \overline{f_2 f_4} + f_3$$

$$= 40.24 \times 10^6 \text{ m/s}$$

For 1/ Active and Passive loop

Adder

1. Half Adder

2. Full Adder

x	y	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\frac{10}{10}$$

$$= 40.24 \times 10^6 \text{ rad/s}$$

Active and Passive Voice

Active voice

He did the work (doer)

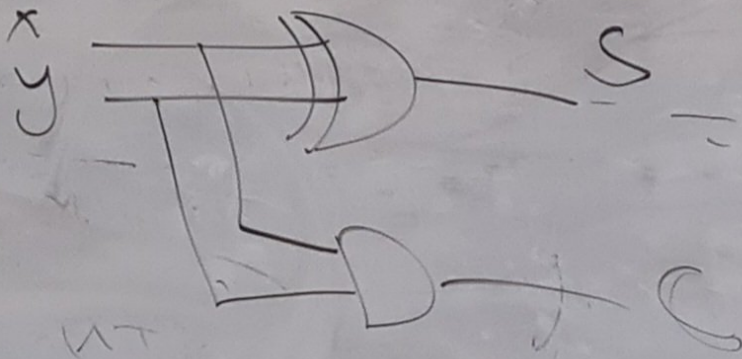
Passive voice

The work was done by him (receiver)

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$$S = \bar{x}y + x\bar{y} = x \oplus y$$

$$C = \underline{xy}$$



Adder

1. Half Adder

2. Full Adder

X	Y	C_i	S	C_{i+1}
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

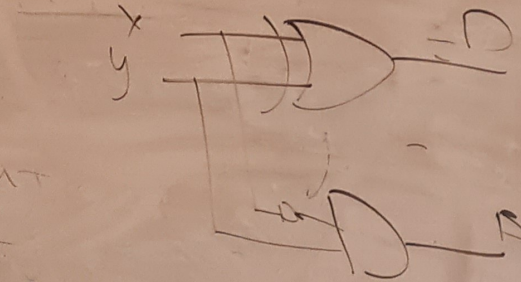
1. Half Subtractor -

2. Full " "

x	y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = \bar{x}y + x\bar{y} = x \oplus y$$

$$B = \bar{x}y$$



$$= 40.24 \times 10^8 \text{ rad/s}$$